ErisML and DEME: A Structure-Preserving, Framework-Pluralist Pipeline for Auditable Machine Ethics

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Abstract

AI safety and governance increasingly demand that systems' normative judgments be transparent, testable, and auditable, yet deployed alignment overwhelmingly reduces moral evaluation to a scalar—a reward score, a safety probability, a guardrail pass/fail—which both discards moral structure and silently pre-commits to a single ethical framework while appearing neutral. We present **ErisML**, a structure-preserving compiler that parses natural-language scenarios into a typed, canonically-hashed *moral-graph* intermediate representation and runs a plurality of framework analyses over it (consequentialist, Kantian with an SMT-checked universalizability gate, virtue, care), refusing to aggregate when frameworks disagree; and **DEME** (Democratically-Governed Ethics Modules), an evaluation engine mapping situations to a canonical nine-dimensional moral vector and per-stakeholder tensor, with constitutional modules that hold hard veto. We frame the methodology as *Philosophy Engineering*: converting normative claims into explicit models with declared invariance assumptions, extracting measurable predictions, and testing them. We report reliability (inter-model agreement $\mathrm{ICC}(2,k) = 0.97$ on the harm subspace), a measured safety property (robustness to meaning-preserving manipulation, surfaced as invariance violations), and pluralist governance (a worst-off/escalate policy that routes genuine disagreement to human review). Both systems are open and installable.

1 Introduction

Deployed AI alignment overwhelmingly reduces moral evaluation to a **scalar**: a reward-model score, a safety-classifier probability, a guardrail pass/fail. This is computationally convenient and, we argue, structurally wrong in two ways. First, it *discards moral structure*—who is harmed, under which maxim, with what consent, in tension with whose rights—collapsing a high-dimensional situation into one number from which the reasoning cannot be recovered. Second, and more insidiously, any single score *silently pre-commits to one ethical framework* (typically aggregative consequentialism) while presenting itself as neutral. A system that returns "0.83 safe" has made deep normative choices that are now unauditable.

We take a different stance, by analogy to compilers. A modern compiler does not map source text directly to machine code; it first builds a *typed intermediate representation* that preserves program structure, enabling analysis and verification before any lowering. Moral evaluation deserves the same discipline: preserve the structure of the situation in an inspectable representation, apply explicitly named ethical frameworks to it, and only then—transparently, and revocably—contract to a decision.

We present two open, shipped artifacts that realize this. **ErisML** is a structure-preserving compiler that parses natural-language scenarios into a typed *MoralGraph* intermediate representation (a canonically-hashed DAG over stakeholders, acts, maxims, commitments, facts, and norms) and runs a plurality of framework *projections* over it—consequentialist, Kantian (with an SMT-checked universalizability gate), virtue, and care. Critically, when frameworks disagree, ErisML *refuses to aggregate*, exposing the disagreement as a first-class, caller-resolved metaethical event rather than laundering it into a number. **DEME** (Democratically-Governed Ethics Modules) is the evaluation engine: it maps situations to a canonical nine-dimensional *moral vector*—physical harm, rights, fairness, autonomy, privacy, societal/environmental, virtue/care, legitimacy/trust, and epistemic quality—extended to a per-stakeholder moral *tensor*, with tier-0 constitutional modules that hold hard veto power.

We frame the underlying methodology as *Philosophy Engineering*: the practice of converting normative claims into explicit models with declared invariance assumptions, extracting measurable predictions, testing them against data, and versioning the result. The framework is its existence proof. We report (i) **reliability**—inter-model agreement $\mathrm{ICC}(2, k) = 0.97$ on the harm subspace; (ii) a measured **safety property**—robustness analysis showing where meaning-preserving manipulations (euphemistic rewrites) move opaque moderators across decision thresholds, surfaced here as invariance violations; and (iii) **pluralist governance**—a worst-off/escalate policy that routes genuine cross-stakeholder disagreement to human review instead of overruling it. Both systems are installable with hash-chained audit artifacts.

Contributions. (1) A structure-preserving compiler IR for moral reasoning that survives to the decision boundary; (2) first-class framework *pluralism with honest disagreement*, including an SMT-verified Kantian gate; (3) a canonical nine-axis moral representation with per-stakeholder tensors and constitutional veto; (4) an auditability layer (canonical hashing, decision proofs, an out-of-band monitor whose only failure output is *requires human review*); (5) the articulation of Philosophy Engineering as a falsifiable, benchmarked methodology, with the framework as a worked instance.

2 Related Work

Machine ethics. The field has long been organized around top-down approaches that encode explicit rules and bottom-up approaches that learn moral behavior from data or feedback [Wallach and Allen, 2009, Anderson and Anderson, 2011]. Recent bottom-up systems learn moral judgments at scale—Delphi from a large corpus of human verdicts [Jiang et al., 2021], and classifiers trained on the ETHICS benchmark [Hendrycks et al., 2021]. These advanced what is learnable, but their output is a *descriptive scalar* whose reasoning is not recoverable and whose implicit ethical commitments are not declared. Our framework is neither purely top-down nor purely learned: it compiles the situation into structure and then applies explicitly named frameworks to that structure, so a verdict carries both its grounds and its theoretical commitments.

The scalar-collapse problem has ethical, not merely engineering, roots. Reducing a moral situation to one number is a form of *moral monism*—the assumption that a single currency of value underlies all considerations—and it inherits monism's classic difficulties. Williams' critique of utilitarian aggregation [Williams, 1973] and Ross's account of *prima facie* duties [Ross, 1930] both insist that moral considerations are plural and not losslessly summable. We take this seriously at the level of representation: DEME's nine dimensions are intended as plural, irreducible considerations in the spirit of a Rossian manifold, not as components of a hidden scalar.

Value pluralism and incommensurability. That ErisML *refuses to aggregate* across conflicting frameworks is not an engineering convenience but a commitment to value pluralism and the incommensurability of values [Berlin, 1969, Williams, 1981, Chang, 1997]. When a Kantian and a consequentialist analysis genuinely diverge, silently trading them into a single recommendation manufactures a false commensuration; surfacing the disagreement as a first-class, caller-resolved event is the philosophically honest output. To our knowledge, no deployed alignment system treats inter-framework disagreement as a structured output rather than noise to be averaged away.

Value alignment and constitutional AI. Mainstream alignment embeds values implicitly—in model weights via RLHF [Christiano et al., 2017], or semi-explicitly via a natural-language constitution [Bai et al., 2022]—within a broader project of specifying what alignment to human values means [Gabriel, 2020]. A constitution is a real step toward explicitness, but its application remains an opaque forward pass returning a scalar preference. We make both the framework and its application explicit, inspectable, and, for the Kantian gate, formally checkable.

Pluralistic alignment and computational social choice. Our verdict-level pluralism connects to recent calls for pluralistic alignment [Sorensen et al., 2024] and to the long tradition of computational social choice on preference aggregation [Brandt et al., 2016], including its emerging application to aligning language models [Conitzer et al., 2024]. The distinction we draw is between an *implicit* social-welfare function buried in a reward model and an *explicit, declared* aggregation rule—here, a worst-off/escalate policy whose normative commitments are stated and contestable rather than latent.

Formal and deontic ethics. For hard constraints we build on deontic logic [von Wright, 1951, Horty, 2001], Hohfeld's analysis of fundamental legal relations [Hohfeld, 1917], and prior attempts to formalize the categorical imperative for machines [Powers, 2006, Lindner et al., 2019]. ErisML's SMT-checked universalizability gate and its D_4-structured Hohfeld module sit in this lineage. But we resist the assumption that *all* moral structure is axiomatizable: the graded dimensions are grounded empirically, closer in spirit to moral particularism's holism of reasons [Dancy, 2004] than to a complete deductive system—a deliberate hybrid of verified hard constraints and learned soft structure.

Governance and auditability. Documentation and audit frameworks specify *what* should be disclosed about a system [Mitchell et al., 2019, Raji et al., 2020, National Institute of Standards and Technology, 2023], but stop short of a mechanism that *produces* inspectable moral reasoning. ErisML's canonically-hashed moral graphs, hash-chained decision proofs, and review-only monitor are a concrete substrate to which such governance can attach—turning auditability from a reporting practice into a property of the computation.

Position. No prior line combines (i) a structure-preserving intermediate representation that survives to the decision boundary, (ii) first-class framework pluralism with honest, non-aggregated disagreement, (iii) an executable and partly formally-verified artifact, and (iv) an auditable decision trail—grounded throughout in plural-duty and incommensurability ethics rather than in a single value currency. The dimensional structure and the descriptive-invariance properties our representation is built to satisfy are developed in separate foundational work [Bond, 2026c]; here we treat them purely as engineering desiderata and evaluate them empirically, making no claim that depends on that apparatus.

3 ErisML: a structure-preserving moral compiler

ErisML compiles a natural-language scenario into a typed *MoralGraph* intermediate representation and runs a plurality of ethical analyses over it. Ingestion proceeds through a multi-pass pipeline: the text is segmented and passed to tiered extractors—deterministic rule-based, LLM-backed (with a critic pass), and a calibrated LaBSE probe—which populate entities, stakeholders, acts, norms, and *ethical facts*. These are resolved to a canonical form and *promoted* to the MoralGraph: a directed acyclic graph over six node kinds (stakeholder, act, maxim, commitment, fact, norm) and eleven edge kinds (e.g., `performs`, `imposes_on`, `consents_to`, `would_violate_if_universalised`), fingerprinted by a canonical SHA-256 hash so that any two runs producing the same moral structure are provably identical.

Over this substrate, ErisML runs four framework *projections*—consequentialist, deontic (Kantian), virtue, and care—each an independent analyzer emitting a `ProjectionResult`: a native verdict, a normalized polarity in {permit, forbid, escalate, neutral}, and categorical *gate findings*. The deontic projection is notable: universalizability is checked by an SMT solver (Z3), and violations of the mere-means and valid-consent gates are reported as discrete findings rather than folded into a score. When two or more projections disagree in polarity, ErisML populates a `cross_projection_disagreement` field and *declines to aggregate*—the choice between conflicting ethical frameworks is surfaced to the caller as an explicit metaethical act, not silently resolved. An out-of-band I-EIP monitor inspects the process along text, activation, and delta lenses; its only sanctioned output on failure is `requires_human_review`. Every run emits a hash-chained *DecisionProof* linking verdict to graph identity and module weights.

4 DEME: Democratically-Governed Ethics Modules

Where ErisML supplies structure and pluralism, DEME supplies the graded, multi-dimensional valuation. DEME evaluates a situation—represented as per-party *ethical facts*—through a layered ensemble of *Ethics Modules* (EMs): a fast reflex layer of hard vetoes, then tactical and strategic layers, with every registered EM emitting a contribution to a canonical nine-dimensional *moral vector*: physical harm, rights, fairness, autonomy, privacy, societal/environmental, virtue/care, legitimacy/trust, and epistemic quality (each in $[0, 1]$; harm inverted). For multi-stakeholder situations this lifts to a *moral tensor* indexed by dimension and party. Verdicts take one of five values—{strongly_prefer, prefer, neutral, avoid, forbid}—computed from the vector under explicit thresholds, with any veto forcing `forbid`.

EMs are organized into governed *tiers*: tier-0 *constitutional* modules (the *Geneva* module, encoding Geneva-Convention and fundamental-rights constraints, with non-disableable hard veto), then core-safety, rights/fairness, soft-value, and meta-governance tiers. A Hohfeld module represents the four fundamental normative positions with D_4 dihedral-group structure, making correlative relations (claim↔duty, liberty↔no-claim) explicit and testable. Module *weights* are not fixed: they are set by *ethos profiles*—the "democratically-governed" element—calibrated from a normative corpus, so that distinct stakeholder value systems instantiate distinct, declared evaluators rather than one universal scorer. Per-party verdicts aggregate conservatively (most-restrictive-wins).

Why these dimensions. The nine axes are derived, not stipulated. They arise from crossing two independent classifications of a moral consideration—its *scope* (individual, relational, collective) and its *mode* (what matters, who decides, what is known)—yielding a 3×3 structure whose nine cells are the dimensions. The construction is validated by *subsumption*: the principles of major applied-ethics frameworks map onto the nine dimensions without residue—the EU High-Level Expert Group's seven requirements for trustworthy AI [High-Level Expert Group on Artificial Intelligence, 2019], Beauchamp and Childress's four principles of biomedical ethics [Beauchamp and Childress, 2019], and the Markkula Center's six ethical lenses [Markkula Center for Applied Ethics] each decompose into the nine without a leftover consideration. The completeness claim is explicitly falsifiable: exhibit a genuine moral consideration that is independent of all nine and not a combination of them, and the basis must be extended. We therefore treat the nine-dimensional representation as *provisionally sufficient and empirically revisable*—a declared modeling choice in the sense of Philosophy Engineering (§6), not a claim that these are the unique axes of value. The derivation, the subsumption arguments, and cross-lingual stability evidence are developed in the foundational work [Bond, 2026c]; we use the result here as an engineering basis and evaluate it empirically.

Instruments. The framework's commitments are *structural*: that moral evaluation be *multi-dimensional* (preserving the trade-offs a scalar destroys), *invariant* under meaning-preserving re-description, and *projectable* onto task-specific bases. A task may call for a coarser or finer basis, and the framework treats such variants as declared *instruments*—projections or refinements of the canonical nine (§4.1). The reliability study (§5) uses a seven-axis harm space that resolves harm into physical, emotional, and financial channels; the governance study uses a ten-module evaluator that adds task-salient axes (fidelity, externality, repair). These are variations on one theme, related to the canonical basis by the projection map of Appendix A, not competing ontologies.

4.1 Instruments, the moral core, and invariance

We make the preceding precise without committing to a privileged basis.

Instruments. A *moral instrument* is a pair (μ, ν) of an evaluation map $\mu : \mathcal{S} \to \mathbb{R}^d$ on situations $\mathcal{S}$ and an order-preserving *verdict map* $\nu : \mathbb{R}^d \to \mathcal{V}$ into the totally ordered verdict set $\mathcal{V} = (\mathsf{forbid} < \mathsf{avoid} < \mathsf{neutral} < \mathsf{prefer} < \mathsf{strongly\_prefer})$. The canonical DEME instrument has $d = 9$.

Commensurability and the moral core. Two instruments (μ_k, ν_k) and (μ_l, ν_l) are *commensurable* if their spaces share a common linear quotient—a *moral core* $C \cong \mathbb{R}^c$ with surjections $\pi_k : \mathbb{R}^{d_k} \to C$ and $\pi_l : \mathbb{R}^{d_l} \to C$ such that $\pi_k \circ \mu_k = \pi_l \circ \mu_l$ on shared situations. The core carries the value information the instruments agree on; each kernel $\ker \pi_k$ carries task-specific structure the other does not resolve. This is the precise sense in which the basis is "arbitrary": it is fixed only up to the choice of instrument, while the core and the verdict order are shared. The canonical 9-axis instrument, the 7-axis harm space, and the 10-module evaluator are pairwise commensurable through a c-dimensional core (Appendix A).

Invariance. Let $\mathcal{T}$ be a set of *meaning-preserving* re-descriptions of situations (paraphrase, reframing, reordering). An instrument *respects invariance to tolerance* ϵ if $\|\mu(Ts) - \mu(s)\| \leq \epsilon$ for all $T \in \mathcal{T}$ and $s \in \mathcal{S}$; a *violation* is a transformation that crosses a verdict boundary, $\nu(\mu(Ts)) \neq \nu(\mu(s))$. The manipulation study (§5) measures this violation rate directly. We state invariance as a tolerance over a *set* of transformations—and deliberately *not* as a group action—to keep the claim empirical and minimal.

5 Evaluation

An auditable machine-ethics framework must satisfy three properties: its judgments must be *reproducible*, *robust to meaning-preserving manipulation*, and *honest about disagreement*. We consolidate evidence from three studies; full protocols, data, and reproduction notebooks appear in the companion papers [Bond, 2026b,a], and we report verified figures here rather than re-deriving them.

5.1 Reliability

A six-model panel (qwen3, qwen3-small, glm-5, gpt-oss, minimax-m2, gemma) independently scored 31 scenarios on a seven-dimension harm subspace—a measurement-reliable projection of the canonical representation (§4). Inter-model agreement is high: overall $\mathrm{ICC}(2,k) = 0.969$ with Krippendorff interval-$\alpha = 0.836$ over $n = 217$ rows, and per-dimension ICC ranges from 0.893 (physical, autonomy) to 0.983 (financial). Test–retest reliability across re-runs is Pearson $r = 0.87$–0.98 per model. The framework's moral assessments are reproducible across independent models and repeated evaluation.

5.2 Robustness to manipulation

A system whose verdicts can be flipped by paraphrase that preserves the underlying act is unsafe; descriptive invariance is therefore a core requirement. Probing it across 161 scenario–perturbation pairs, *euphemistic* rewrites—which soften the description while preserving the act—shift verdicts toward the permissive in 13.7% of cases, collapsing `forbid` verdicts from 44 to 27 and reducing assessed harm by 14.0 points on a 0–70 scale (mean ordinal shift +0.65 toward permissive). Symmetrically, dramatizing rewrites shift 16.1% toward the restrictive. Crucially, the framework *surfaces* each such shift as an invariance violation rather than silently re-deciding—the manipulation becomes an auditable event, not an undetected exploit.

5.3 Pluralist governance

Two stakeholder ethos—a harm/care advocate and a fairness/fidelity community, each a declared module-weighting—evaluated 231 real moral dilemmas. The ethos produce systematically different moral vectors (mean L_2 distance 0.140; verdict divergence 34.2%; directional sign-tests $p < 0.001$ on physical-harm and fidelity dimensions), confirming that declared value systems instantiate distinct evaluators rather than one universal scorer. The multi-dimensional representation also separates situation-level harm from agent culpability: aggregate harm is uncorrelated with human blame judgments (Spearman $\rho = -0.05$, n.s.), while the fairness and care dimensions track it ($\rho = -0.18$, $p = 0.006$; $\rho = -0.16$, $p = 0.014$)—structure a scalar score discards. Under the worst-off/escalate policy, the governed decision differs from the most-permissive single stakeholder in 33.8% of cases, and every dilemma exhibits ≥ 2 distinct framework polarities, routing genuine disagreement to human review.

5.4 External validation against human moral foundations

The preceding results concern the framework's internal behavior; a sharper test is whether its dimensions recover *independently human-annotated* moral structure. We scored a stratified sample of 280 comments from the Moral Foundations Reddit Corpus [Trager et al., 2022]—hand - labeled by trained annotators for moral foundations—on DEME's dimensions via an LLM panel, and correlated the two. Four pre-registered alignments hold and are each the *argmax* over foundations: `care_protection` with Care ($\rho = 0.51$), `fairness_equity` with Equality ($\rho = 0.48$), `legitimacy_trust` with Authority ($\rho = 0.43$), and `vow_fidelity` with Loyalty ($\rho = 0.45$); all $p < 10^{-15}$, $n = 280$. The result *replicates across two independent model families* (qwen3 and glm-5) with overlapping 95% bootstrap confidence intervals, ruling out a single-model artifact; the Purity foundation, which has no DEME analog, is the quiet negative control. The dimensions are thus not arbitrary coordinates: they track human moral judgment where they should and stay quiet where they should not (companion notebook).

Honest negatives. DEME is not a culpability classifier: on this fully contested corpus, near-universal escalation does not by itself discriminate hard from easy cases. We report this because the framework's contribution is to *surface* disagreement faithfully, not to adjudicate it.

Reproducibility. Every figure in this section is reproducible from released derived data and self-contained notebooks (`numpy/scipy/matplotlib` only); the LLM-panel scoring scripts are provided for re-derivation from public corpora. The released data contain derived scores and human label proportions only—no raw post text.

6 Philosophy Engineering

The framework instantiates a methodology we call *Philosophy Engineering*: the construction of executable, falsifiable implementations of normative frameworks, with modeling assumptions declared, predictions extracted and tested, and the result versioned and revised when predictions fail. It is neither normative ethics (it does not pronounce which theory is correct) nor descriptive moral psychology (it does not merely predict human judgments); it is the practice between them—making normative commitments explicit enough to be built, measured, and corrected.

Four commitments distinguish it. First, *declared modeling choices*: every bridge from a moral claim to a computational object carries its epistemic status—definition, assumption, conditional theorem, or empirical result—so that what is proven is not confused with what is posited. Second, *stated invariances*: the transformations under which an evaluation must be stable (§4.1) are written down and tested, not assumed silently. Third, *extracted predictions*: a model earns its keep by entailing measurable claims—here, inter-model reliability and manipulation-violation rates (§5). Fourth, *versioned revision*: when a prediction fails the model is amended and the change recorded, exactly as the nine-dimensional basis is held under a falsifiable completeness claim (§4) rather than as fixed truth.

ErisML and DEME are a worked instance of all four: the moral-graph IR and the nine-dimensional representation are declared modeling choices with stated assumptions; the invariance principle is a declared invariance whose violations §5 measures; reliability and robustness are extracted predictions tested against data; and both systems ship under versioned APIs whose evaluation history is part of the artifact. The claim we make for Philosophy Engineering here is narrow and by example—that normative frameworks can be built to the standard of explicitness, testability, and revisability we expect of any engineered system; a fuller account is left to separate work.

7 Limitations and Open Problems

The ingestion bottleneck. The pipeline is bounded by its weakest stage. Moral evaluation begins by extracting structured facts from natural language, and an error there propagates to a verdict that is then computed flawlessly and certified with hash-chained certainty—formally verified garbage out. Three mechanisms bound, but do not eliminate, this: a critic pass that re-checks each extraction against the source text, a calibrated probe whose disagreement with the LLM extractor flags low-confidence parses, and the out-of-band monitor, whose only output on anomaly is `requires_human_review`. Each catches some bad parses; none audits the *parser itself*. Bounding how often a subtle maxim or stakeholder is missed, and with what downstream effect on the verdict, is the single most important open problem for the framework. The graded dimensions inherit a further limitation: their grounding corpora are predominantly English-language, Western, and model-mediated, so the figures of §5 are inter-model, not human-grounded.

The escalation–throughput tension. Refusing to aggregate is honest but, at the limit, expensive: under worst-off/escalate roughly 91% of our evaluation cases route to human review, which would make the system a triage tool rather than an autonomous layer. Two considerations temper this. First, the corpus is worst-case: every AITA post *is* a submitted moral dilemma, and notably its "clear" cases escalate at essentially the same rate as its contested ones (90.7% vs. 91.2%), because escalation is driven by genuine disagreement *among the ethical frameworks*—near-universal on hard dilemmas, and near-absent on the benign content that dominates real streams, where the frameworks simply agree to permit. The 91% is thus an upper bound on an all-dilemmas distribution, not an expected deployment rate. Second, worst-off/escalate is the maximally cautious endpoint of a tunable family: escalating only on *decision-relevant* polarity conflicts (permit-vs-forbid rather than differences of degree), gating by stakes or severity, or imposing an escalation budget each trade pluralist purity for throughput—and, in the spirit of the framework, that trade is itself a declared, auditable governance choice rather than a hidden threshold. Characterizing this integrity–throughput frontier is important future work.

Validating the moral core. The commensurability of instruments (§4.1) currently rests on a *proposed* core map (Appendix A). Its validation is concrete and high-priority: run the canonical 9-axis, 7-axis, and 10-module instruments on the *same* inputs and test whether $\pi_k\mu_k \approx \pi_l\mu_l$ on the shared core to a stated tolerance. A first step on effective dimensionality is already encouraging—on the governance corpus the leading principal component of the moral vectors explains only 35% of the variance and five components are needed for 90% (companion notebook), so the representation is empirically far from one-dimensional, and an external check against human moral-foundation annotations (§5) finds each tested dimension aligned with—and most correlated with—its intended foundation. The cross-instrument map is also now supported empirically: scoring the same 280 inputs on the 9-axis and 10-module instruments, their core projections agree at mean $\rho = 0.88$ (per-axis 0.75–0.95), and the differently-based 7-axis harm space recovers the shared subcore ($\rho = 0.51$–0.77); full validation across all instruments and corpora remains future work. The verdict thresholds, the worst-off aggregation, and the constitutional vetoes likewise remain declared normative choices: contestable, not derived.

8 Conclusion

We have presented ErisML and DEME as a structure-preserving, framework-pluralist, and auditable pipeline for machine ethics: a typed moral-graph IR that survives to the decision boundary, plural framework analyses that refuse to silently aggregate, a derived multi-dimensional evaluation with constitutional veto, and a hash-chained audit trail. The framework's judgments are reproducible across models ($\mathrm{ICC}(2,k) = 0.969$), expose meaning-preserving manipulations as invariance violations rather than absorbing them, and route genuine stakeholder disagreement to human review under a declared policy—none of it requiring the collapse of moral structure to a scalar, and all of it open and installable. More broadly, the work is an argument by construction for Philosophy Engineering: that normative commitments are better made explicit, testable, and revisable than left implicit in a number. We release the artifacts in that spirit—as infrastructure to be scrutinized, contested, and extended.

A Instrument-to-core projection map

Table 1 gives the proposed correspondence underlying §4.1. The shared moral core C comprises the seven axes on which all three instruments agree (up to coarsening). Beyond the core, the canonical instrument adds *privacy* and *societal/environmental*; the ten-module evaluator adds *fidelity*, *externality*, and *repair*; and the seven-axis harm space resolves the harm axis into *physical*, *emotional*, and *financial* channels and adds a *social/identity* pair. These extensions lie in the respective kernels $\ker \pi_k$; commensurability is claimed on C only, and the extensions are *not* asserted to be inter-derivable. (Proposed map; axis assignments to be confirmed against the released instrument specifications.)

Core axis c_i	Canonical (9)	Harm space (7)	Module eval. (10)
Harm	`physical_harm`	physical†	harm
Rights	`rights_respect`	(identity)	rights
Fairness	`fairness_equity`	—	fairness
Autonomy	`autonomy_respect`	autonomy	autonomy_consent
Legitimacy/Trust	`legitimacy_trust`	trust	legitimacy_trust
Epistemic	`epistemic_quality`	—	epistemic_quality
Care	`virtue_care`	—	care_protection

Table 1: Proposed instrument-to-core map. †: the harm space refines this axis into physical/emotional/financial channels. "—": the harm space, being harm-focused, does not resolve this core axis. Instrument-specific extensions (privacy, societal; fidelity, externality, repair) are omitted as they lie outside C.

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